Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS55]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

PARALLEL COMPUTING SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Explain in detail about scope of parallel computing.	**07**
	b.	Limitations of Memory System Performance.	**07**
2.	**a.**	Demonstrate the Decomposition Techniques with an example	**07**
	b.	Demonstrate the different Mapping methods with an example.	**07**
3.	**a.**	Discuss the Principles of Message Passing Programming.	**14**
		OR	
4.	**a.**	Explain the Collective Communication and Computation Operations and Groups and Communicators.	**14**
5.	**a.**	Define threads? Why threads are needed.	**06**
	b.	Explain the Thread Creation and Termination concepts with example.	**08**
		OR	
6.	**a.**	Explain the Composite Synchronization Constructs with example.	**08**
	b.	Discuss the ideas for Designing Asynchronous Programs.	**06**
7.	**a.**	Explain OpenMP programming model with sample OpenMP program?	**07**
	b.	Listout the Environment variables in OpenMP.	**07**
		OR	
8.	**a.**	List the Issues in Sorting on Parallel Computers.	**04**
	b.	Discuss the Minimum Spanning Tree: Prim's Algorithm for finding shortest vector value.	**10**